The Hand of Corporate Management in Capital Allocations

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Abstract

One of Chandler's basic insights was that the new organizational structures adopted after the 1920s reflected a specialization of the roles of business and corporate managers. Business managers coordinated functions and corporate managers allocated resources. This article studies the impact of corporate management on capital allocation decisions by comparing investment behavior in multi- and single-business companies. Using a new taxonomy to control for the quality of each business unit, we analyze a cross-sectional sample of U.S. companies and identify a number of empirical regularities that highlight the allocative decisions of corporate management. In particular, we find that, when dealing with cash-needy businesses, multi-business firms invest more intensively in those that are less profitable. We find no evidence that this subsidy results in a higher success rate in making those businesses profitable over time. We suggest a number of explanations for this empirical pattern.

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Introduction

As Alfred Chandler put it, starting in mid-19th century, "the visible hand of management replaced the invisible hand of market forces" (Chandler, 1977), thus making the existence of a managerial hierarchy one of the defining characteristics of the modern business enterprise.

Chandler explained the change from the entrepreneurial to the managerial enterprise on the basis that the increasing scale of operations in the modern firm created

gains to control and coordination by management. Studying that proposition immediately became a fundamental challenge to business scholars. From early work on the multi-divisional form (e.g. Williamson, 1975) and diversification (e.g. Rumelt, 1974) to many recent developments in the business and economics field, a mixed picture of what "the hand of management" can contribute to the modern corporation has emerged. To many instances of Chandler's basic argument about the benefits of managerial coordination, the literature has posed various examples of the limits to that coordination (limits of which Chandler was well aware). Scholars have identified misaligned incentives (Jensen, 1986) and managerial bounded rationality (Simon, 1947) as two of the factors that play a larger role in increasing the costs of managerial coordination.

In Chandler's view, the need to coordinate an always-increasing number of units and activities within any given firm not only required the emergence of a professional managerial class but also the creation of a number of organizational processes that have become basic staples of the modern corporation. In particular, top management "took the place of the market in allocating resources for future production and distribution" (Chandler, 1977) and so the modern capital budgeting process was born.

The operation of the budgeting process-the internal capital market-is a key process in a multi-business firm. Corporate and divisional managers must decide how to distribute investment capital among the company's different business units and/or the projects within those units. This article studies the role that Chandler's "hand of management" plays in the capital allocation efficiency of multi-business firms. To that effect, we study the investment behavior of multi-business firms over a 16-year period (1989-2004) and compare it with that of single-business firms. We also develop a taxonomy of businesses that allows us to separate them into four categories depending on their profitability as well as the balance between their ability to generate cash and their need for cash. In this analysis, we provide evidence of what "the hand of management" does to investment behavior in multi-business firms.

In studying the capital allocation decisions of management across firms it is necessary to create a proxy measure for the investment opportunities presented by a business unit-a characteristic we will term "quality". A common way to measure the quality of a business unit is to use an estimate of its Tobin's Q . For example, Scharfstein (1998) observes that corporations seem to overinvest in low-Q (low-quality) businesses and underinvest in high-Q (high-quality) businesses. [1] In this article, we advance a new methodology to categorize the quality of a business as an investment opportunity. This categorization is based on two simple characteristics of a business, namely, its growth and its profitability. Using these two variables has a number of advantages relative to Tobin's Q .

[1] The main problem with this methodology is that Tobin's Q is not always a good estimate of the future value of a business. Generally, cross-subsidization studies have estimated the value of a business unit's investment opportunities by taking the median Q for all the stand-alone firms in the target business unit's industry. However, as Whited (2001) notes, it is unclear whether standalone firms have access to the same set of investment opportunities than corporations. If multibusiness firms possessed an advantage in their access to better business opportunities (e.g. due to

First, Tobin's Q relies on a measure of a business' market value which is unobservable for business units of multi-business firms. By contrast, the growth rate and profitability of a business unit are directly observable in the COMPUSTAT segment database. Second, using two variables instead of one to categorize businesses allows for a richer analysis. Thus, in our classification we distinguish among four different types of businesses (self-sufficient/not-self-sufficient business vs. high/low profitability) whereas the previous literature distinguishes only between two types (high/low Tobin's Q). We believe the distinction between self-sufficient businesses (those that generate enough cash flow to cover their own investment capital requirements) and cash-needy ones (those that have to "borrow" capital from corporate headquarters) allows for a more nuanced assessment of the reasons corporate management might have for shifting capital across business units. Corporations have the option of using capital generated in other parts of the firm to fund a particular business unit whereas stand-alone firms have to rely exclusively on external investors. The constraints and requirements on corporate capital are less strict than the ones on external capital, thus making it easier for multi-business firms to invest in cash-needy businesses.

Internal Capital Markets

The definitive descriptive model on the operation of internal capital markets from a corporate strategy perspective was proposed by Bower (1970). The basic resource allocation process (RAP) model proposed by Bower involves multiple organizational levels, as operational managers within the divisions provide the definition of the several investments available to the company to their divisional managers, who in turn provide the impetus to sponsor some of those investments in budget meetings with corporate headquarters. Finally, corporate headquarters decides on the final allocations while also being in charge of providing the structural context (organizational design, incentives, etc.) in which this process takes place. In Bower's model, the internal capital market actually functions as a "market" for reputation, with their average superior resources), then the market values of stand-alones would systematically underestimate the value of businesses in corporations, thus affecting Tobin's Q value as a benchmark. Under these circumstances, the finding in the literature that segment investment is less responsive to segment estimated Tobin's Q in corporations than it is in stand-alones would not be a consequence of investment inefficiencies but a consequence of a systematic measurement bias. Moreover, Chevalier (2004) also points out that the use of industry Q as a proxy for segment Q is further complicated if the activities of the businesses within a diversified corporation are somehow related. impetus-producing managers given the power to allocate capital as long as their track records remain sound. Thus, Bower's model provides a picture of the complex mechanisms by which corporations allocate capital as well as the role that corporate management plays in this process.

Further applications of Bower's model have included business development (Noda and Bower, 1996), response to disruptive technology (Christensen and Bower, 1996) and internal corporate

venturing (Burgelman, 1983), among others. One important feature of the RAP model is in the limits it sets to the influence of corporate management in the final allocations. Even though the final decision is taken at the top, the process of selecting prospective investments is mostly determined at the divisional level. Some authors have identified particular situations where it might be advisable to increase the role of corporate management in the decision. For example, Sull (1997) notes that the role of headquarters is especially important in promoting disinvestment, because divisional managers often perceive such disinvestment as jeopardizing their job and compensation and, hence, refuse to support it. Eisenmann and Bower (2000) suggest that when facing large upfront investments in turbulent new markets, the company might be better off if headquarters takes on a larger role in the allocation decision.

As Gilbert and Bower (2005) note, the RAP model's basic perspective comes close to the agency-based explanations of the corporate finance literature that we cited above. In essence, both perspectives present the resource allocation process as a competition among business units for the limited pool of resources available to the corporation (Burgelman, 1991). Other strategy scholars echo this view. Hoskisson et al. (1991) hypothesize that divisional managers are increasingly risk-averse as their firm becomes more diversified, due to the difficulty of headquarters to monitor effectively the proposal and execution of ambitious goals for each division as well as the divisional managers' low incentive to pursue them. Moreover, Hoskisson et al. (1994) find that higher equity stakes among board members makes firms more effective in divestment situations.

From a finance perspective, internal capital markets have the capacity to add value to the firm but can also be a source of problems. For example, a corporation might be more capable to raise money for one of its business units than that unit would be if it were a stand-alone firm. The few empirical studies addressing this "more-money" effect suggest that it is not a major issue (Berger and Ofek, 1995; Comment and Jarrell, 1995). Or, corporations might experience a "smart-money" effect to the extent that corporate managers can do a better job than the external capital markets at shifting funds from one business to another. For example, Stein (1997) argues that, in a situation where external funds are constrained, corporations are in a better position than market investors to gather information about their business units and to identify the businesses with the best prospects. Moreover, they can then use cash generated in other units to fund those "winners". Therefore, if corporate managers can pick the best prospects correctly, then having an internal capital market adds value to a collection of business units even when the total amount of external capital raised by the corporation is the same as it would be for those businesses operating independently. On the other hand, the literature also provides a number of examples that show that corporations may not always be efficient at picking the best investment options, as misaligned managerial incentives (Scharfstein and Stein, 2000) and asymmetric information between headquarters and divisional managers (e.g. Harris and Raviv, 1996) can lead to poor allocations. With some theorists pointing to the left and others to the right, one must turn to the data to seek a resolution of the issue.

One approach to this problem has been to compare the investment behavior of multi-business

firms with that of single-business firms (e.g. Lang and Stulz, 1994; Berger and Ofek, 1995; Scharfstein, 1998). For instance, Scharfstein (1998) found that businesses in poorly performing industries received more investment by a multi-business firm than their stand-alone counterparts. Conversely, divisions of multi-business firms in well-performing industries received less investment than their stand-alone counterparts. This phenomenon, dubbed "corporate socialism," points at a contrast between the behavior of multi- and single-business firms.

Interestingly, Lamont (1997), in a study of oil companies, showed that an exogenous shock to one of the divisions (such as a drop in oil prices) affected the capital allocations in unrelated divisions within the same firm. In particular, Lamont found that following the oil price drop in 1986 companies with oil businesses reduced their investment in non-oil divisions, even though the market conditions for those non-oil businesses had not changed. Thus, Scharfstein and Lamont's studies offer strong evidence of a correlation among the capital allocations to the divisions in a firm although they are less conclusive in showing that such correlation systematically leads to inefficient allocations. Some recent literature has found methodological problems in those findings, thus weakening their validity. For instance, Villalonga (2004), using a new definition of business units, found allocation inefficiencies only in unrelatedly-diversified firms but not in relatedly diversified firms. Other authors have found a similar distinction in the optimality of capital allocations in unrelatedly and relatedly-diversified firms (Rumelt, 1974; Khanna and Tice, 2001).

Data

The COMPUSTAT segment files include data on sales, operating profits, depreciation, capital expenditures and total assets at the business segment level. All publicly listed firms are required by SEC regulations to report those figures for each one of their segments. Then, COMPUSTAT assigns a primary and secondary four-digit Standard Industrial Classification (SIC) code to each segments as well as a "segment name" as reported by the company in order to identify the main activity of each business. One limitation of COMPUSTAT segment data is the different criteria used by firms in deciding what constitutes a business segment. Sometimes, the same firm might report the same business segments differently over different years. To minimize the impact of this problem in our sample, we aggregate the reported segments using 3-digit level SIC industry groups, a common approach in the corporate finance literature (e.g. Lamont, 1997; Ozbas and Scharfstein, 2009). Thus, in our sample, a firm has as many different business units as it has different three-digit codes. This aggregation reduces the noise inherent in the reporting of business segments in the COMPUSTAT files and provides a better picture of the degree of diversification in a company. For example, if a firm reports three different segments with the same three-digit code, it would be misleading to say that such company is diversified, given that all those segments operate in essentially the same line of business. [2]

It is also important to note that a transition toward a new industry identification system started

after 1998. Thus, COMPUSTAT stopped assigning SIC codes for many of the segments in its files and began assigning NAICS codes. Since our sample contains observations from both before and after 1998 we identified equivalent industries in both systems and translated NAICS codes into SIC ones. The raw COMPUSTAT database from 1989 to 2004 contained 105,424 segment-year observations. After consolidating at the three-digit SIC code level and eliminating the segments with incomplete data, 39,452 segment-year observations remained. Table 1 provides a breakdown of the segments in the sample.

Investment Patterns

In order to measure investment behavior one needs to control for the type of business that is the target of that investment. For instance, a business that is starting up requires a large investment upfront, often not justified by the initial stream of cash flows. Conversely, a mature business that has been yielding high returns for a long time and is now shrinking generally requires little investment. This logic should lead firms to systematic differences in investment behavior depending on the type of business that they own.

Most of the previous literature on internal capital markets has used Tobin's Q[3] as a proxy for "quality". The idea, originated by Tobin, is that market values in excess of Table 1 Number of observations in the sample by year

Number of segments in the firm	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004
1	1077	1108	1147	1123	1156	1147	1194	1236	1228	1176	1138	1154	1097	1123	1197	1128
2	467	601	438	445	427	455	436	396	383	415	460	558	586	687	730	650
3	415	407	421	386	405	430	412	396	361	410	420	426	350	452	438	401
4	239	236	215	224	232	215	198	178	164	184	195	206	193	214	228	236
5	84	102	98	86	81	75	63	59	62	74	88	93	91	97	110	116
6	31	45	41	37	42	42	37	28	39	31	46	36	41	43	49	42
7	25	23	28	30	37	31	34	38	41	46	44	51	35	37	33	42
8	8	7	3	9	7	6	8	7	5	9	13	10	9	4	0	0
9	4	3	10	11	11	18	8	8	0	27	5	0	5	13	19	12
10	18	18	12	12	12	12	17	10	10	9	8	9	7	5	0	0
Number of observations per year	2368	2550	2413	2363	2410	2431	2407	2356	2293	2381	2417	2543	2414	2675	2804	2627

book values signal expected future profits. (The obvious problem with that logic is that book values leave out intangible capital, so that a high Tobin's Q may simply signal the presence of intangibles rather than opportunity.)

[2] Because of this three-digit code aggregation we use the terms "multi-business" and "diversified" equivalently throughout this paper. Similarly, we use the terms "single-business" and "focused" equivalently throughout this paper. It is also worth noting that aggregating business units using SIC codes is common in other segment-based studies of capital investment (e.g. Lamont, 1997). Some studies even use higher levels of aggregation (two-digit level) to consolidate segments (e.g. Scharfstein, 1998; Shin and Stulz, 1998). [3] Tobin's Q is a standard proxy for the quality of a firm's investment prospects and it is generally calculated as the ratio between the market value of a firm and the book value of its assets. In our

Our own approach classifies a business unit by its past growth rate and its profitability, two measures that are observable to both the researcher and management. Specifically, we separate businesses into profitable and unprofitable ones by comparing the ratio of operating profit to assets R to an established "cost of capital" threshold (k). Thus, a business with a return rate R higher (lower) than k is considered profitable (unprofitable).

To measure the return rate (R), we divided business unit operating profits over business unit assets (a variation on the Return over Assets ratio). [4] Regarding the cost of capital threshold, we used a single average cost of capital of 9% as the profitability threshold for all the business units in our sample. [5]

The second dimension in our taxonomy, growth, helps us separate businesses into "cash-needy" and "self-sufficient." If a business grows its assets at rate g per year, it needs to earn at least that much profit, as a fraction of assets, to be self financing. Thus, a cash-needy business is one where $g > R$ and a self-sufficient business is one where $g < R$.[6]

To measure growth (g) we fitted an exponential curve to a moving window of 5-year business sales figures and took the slope coefficient of that fit as the business

unit growth rate for the final year of each 5 -year window. [7] A 2×2 matrix (Figure 1) shows the four discrete types of business that result from this taxonomy. [8]

Businesses in the Cash-Needy Unprofitable (CNU from now on) region in the matrix are not profitable ($R < k$) and cash-needy ($g > R$). In the real world, one can typically find these two characteristics in young "growth" businesses that are in the initial stages of their lifecycle. Corporations and individual investors might allocate capital to those businesses because they expect long-run returns even though their first years might be characterized by low ones. For example, in our sample, one of the business units located in the CNU region is Medco Research Inc., a pharmaceutical company focused on the cardiovascular drug market. This company experienced a 50% average growth rate in the first 5 years. Its average profitability rate over the same period was -15%, although after 7 years the company started turning positive profits. This pattern is not uncommon in businesses in the pharmaceutical industry, which requires large upfront investment

study, we follow Kaplan and Zingales (1997) and we calculate stand-alone firm's Q as MarketValue/ (0.9* BookValue +0.1 MarketValue), where the book value of assets equals Compustat item 6 and the market value of assets equals the book value of assets plus the market value of common equity less the book value of common equity (item 60) and balance sheet deferred taxes (item 74). This simple market to book ratio differs from the standard definition of Q in that it does not estimate the replacement value of assets nor adjusts for taxes. Previous studies have shown that these adjustments are not essential (Perfect & Wiles, 1994). Estimating Tobin's Q for multi-business firm segments using industry medians based on single-business firms is a standard practice in the literature (e.g. Rajan et al., 2000; Ozbas and Scharfstein, 2009) [4] We recognize that Return over Invested Capital might be a better measure for our analysis. However, the COMPUSTAT segment files do not include segment Investment Capital data. In the end, and given the widespread use of the return over assets ratio in the finance literature, we used total segment assets as the denominator in our return measure. [5] We acknowledge that the cost of capital varies from firm to firm and even among business units within the same firm. However, the general cost of capital figure used in real firms usually falls within the $8-12\%$ range. We tested the sensitivity of the results in this paper to different values within that range. Our findings were not qualitatively affected by those changes in the cost of capital rate. [6] This is an assumption that might have different validity across industries. Generally speaking, the correlation between sales and assets will be stronger in manufacturing industries and weaker in service and high-technology ones.

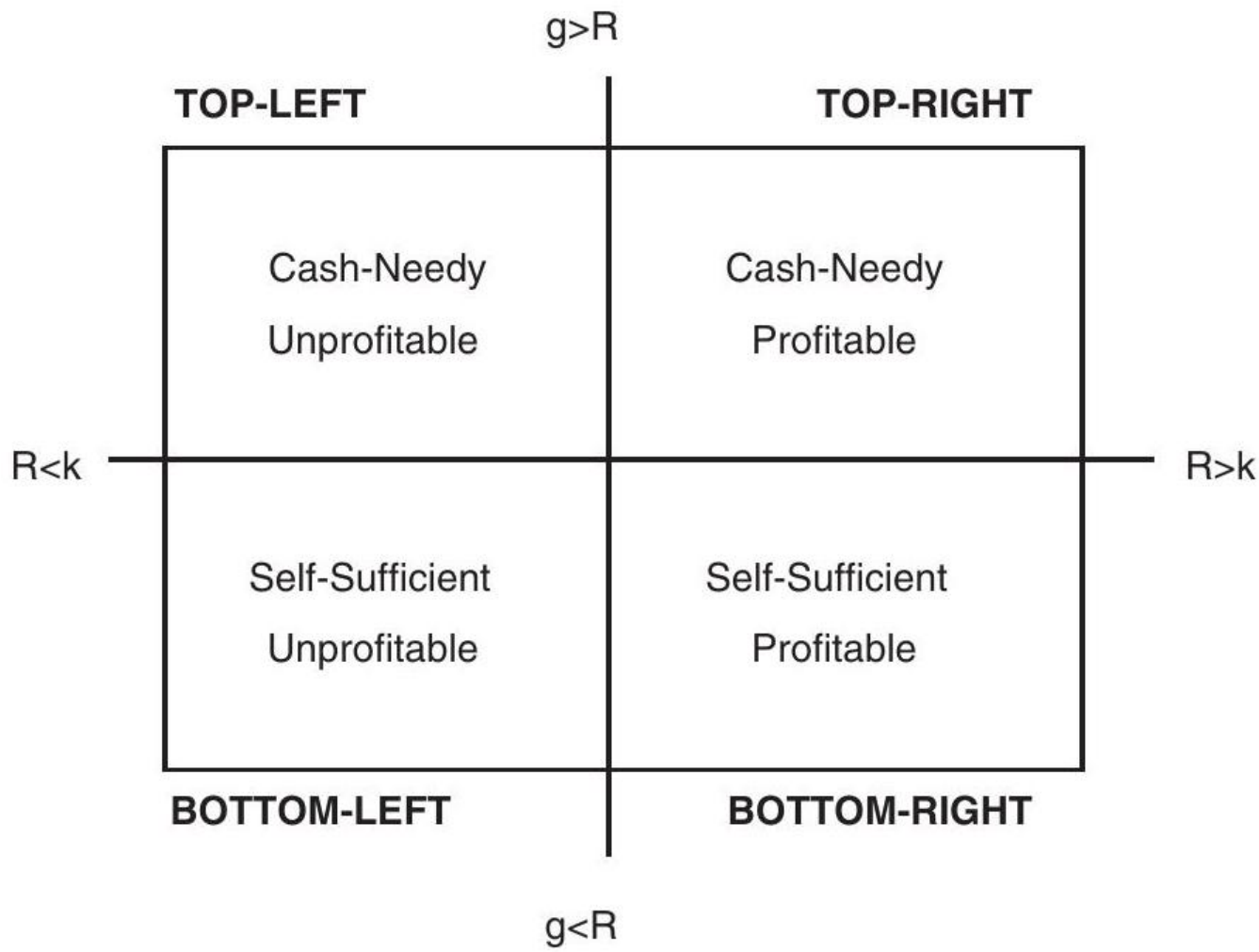

Figure 1 The g-R matrix.

for drug development. However, it is important to note that not all businesses located in the CNU region are "seeds". For example, the distribution business of International Paper Co. presents an average profitability rate around 6% over an 11-year period (1989-1999) and an average growth rate around 10% over that same period. Although its growth rate exceeded its ability to generate cash, it is a mature business. Not exactly a "seed," such businesses may actually be "weeds," absorbing capital from the other business units at International Paper. Therefore, the more this business grows the more firm resources it absorbs, without even returning the cost of those resources (the cost of capital k).

Businesses located in the Cash-Needy Profitable (CNP from now on) region are cash-needy ($g > R$) and profitable ($R > k$). For example, the Oil & Gas division at Apache Corp presents an

[7] Given the endogeneity between investment and asset growth we use sales to measure firm growth. Sales growth is also the standard measure of growth for managers. [8] As with any taxonomy of businesses (e.g. the market-growth market-share BCG matrix) there are limitations in classifying segments into discrete groups. Our classification is based on the relationship between business growth and profitability described in the main text. However, we ran several sensitivity analyses to make sure that our grouping was not spurious.

average profitability around 14% from 2001 to 2004, as well as an average growth rate of around 20% per year. Interestingly, this business had a similar growth rate but lower profitability (6%) in the previous 5 years. Thus, this business can be qualified as a high growth business that delivers low profitability at the beginning but improves after a number of years.

Businesses located in the Self-Sufficient Profitable (SSP from now on) region typically generate more cash than required for their investment needs ($g < R$) and they are profitable ($R > k$). Obviously, that extra cash can be used to invest in other units in the firm. A typical example is the Aircraft Engine division at General Electric Co. This business unit has delivered high profitability rates for a long time (around 20% a year from 1989 to 2004) although its growth rate has stayed below 10% for the most part of that period.

Finally, businesses located in the Self-Sufficient Unprofitable (SSU from now on) region are characterized by a low profitability rate ($R < k$) but self-sufficiency regarding their investment needs ($g < R$). Their low profitability makes these businesses look like a poor investment. At the same time, given their self-sufficiency, these businesses can survive a long time on their own without absorbing external resources from either the corporation or outside investors. For example, Canadian Pacific Railway delivered an average growth of zero between 1990 and 1998, together with an average profitability around 3%. In our sample a good number of businesses belonging to utility companies (railroad, water, electricity, etc.) and commodity producers (steel, food, etc.) are located in this region.

In order to reveal patterns of investment behavior, we estimate a multi-linear model that has business unit investment as the dependent variable. We measure investment as the ratio of business unit capital expenditures to business unit assets. This ratio represents how much money "per dollar of assets" a business unit receives. This way, we automatically control for the natural effect of divisional size on the absolute amount of capital that a business receives. Our additional con- trol variables were the median rate of investment in the business unit's industry and the past growth and profitability of the business unit.

Industry-level median investment should constitute a strong predictor of investment as well as a good control for industry-specific investment cycles. Moreover, the median level of investment in an industry may also be a proxy for the investment opportunities available in it. For instance, an industry-wide high level of investment could be a reaction to good businesses opportunities in that industry.

More specifically, the dependent variable (CAPINT) is net capital expenditures (operating profit minus depreciation) divided by lagged business unit assets. Business unit growth (SEG_G) is the coefficient of the exponential curve fitting segment sales from year $t-5$ to year $t-1$. Business unit profitability (SEG_PROF) is the lagged 5-year average of operating profit minus the cost of assets normalized by sales. IND_CAPINT (the median level of investment in each business unit's industry) is the median of the dependent variable in each business unit's industry. We expect that investment intensity will be positively related to the controls IND_CAPINT, SEG_G and

SEG_PROF.

Our non-control independent variables were dummy variables drawn from the four regions of the g-R matrix. Thus, the variable CASHNEEDY equals 1 for the business units located on the CNU and CNP regions; the variable UNPROF equals 1 for the business units located on the CNU and SSU regions; and CNU equals 1 for the business units in the CNU region. We define LND3 as the logarithm of the total number of business units that coexist in the target business' firm. Thus, LND3 is zero for any target business that belongs to a stand-alone firm and takes a positive value for any target business that belongs to a multi-business firm.

Results

Should we expect any systematic differences between single- and multi-business firms regarding their investment behavior in each one of these four regions? Since some theory points towards internal capital markets being "better" and some points towards inefficiencies, our basic aim in this article is to reveal the actual empirical patterns in the investment behavior of multi-business firms. Hence, our working "hypothesis" is the null-that there will be no significant differences between investment behavior between multi-business and single-business corporations.

To begin, Figure 2 displays the distribution of assets among the four regions in the g-R matrix for both multi-business and single-business corporations. [9] There is a significant difference between diversified and focused firms in the percentage of assets located in the CNU region. Over 15 years, 42.6% of the assets belonging to multi-business firms were located in that region, whereas that same figure for single-business firms was only 29.3%($t = 6.5$). This phenomenon was fairly consist- ent over time, as most of the years in the sample presented a similar pattern. Figure 2 also shows that single-business firms in our sample had proportionately more assets in the CNP region (relative to multi-business firms.) In this case, 17.8% of the assets

of single-business firms were located in that region, whereas that same figure for multi-business firms is only 8.5%($t = 5.7$).

Table 2 (Models 1 and 2) shows the regression results. As expected, the controls load with positive coefficients. The two categorization dummy variables load as one would expect from simple intuition: Unprofitable business units (UNPROF) receive lower rates of investment ($t = -9.81$) and fast growing business units (CASHNEEDY) receive more ($t = 2.72$). The interaction between these two-the CNU region-has a net negative effect on investment intensity ($t = -3.18$).

The "hand of management" is revealed by the interactions with the LND3 variable, which measures the presence and number of sister-business units squabbling over capital. The regression results show that internal competition among business units (LND3) does the following:

[9] As has become traditional in this type of study, we excluded all financial segments (SIC codes 6000 to 6999) because their measures of "assets" differ fundamentally from industrial and retail segments. However, we ran similar tests including those segments in the sample. The results were similar.

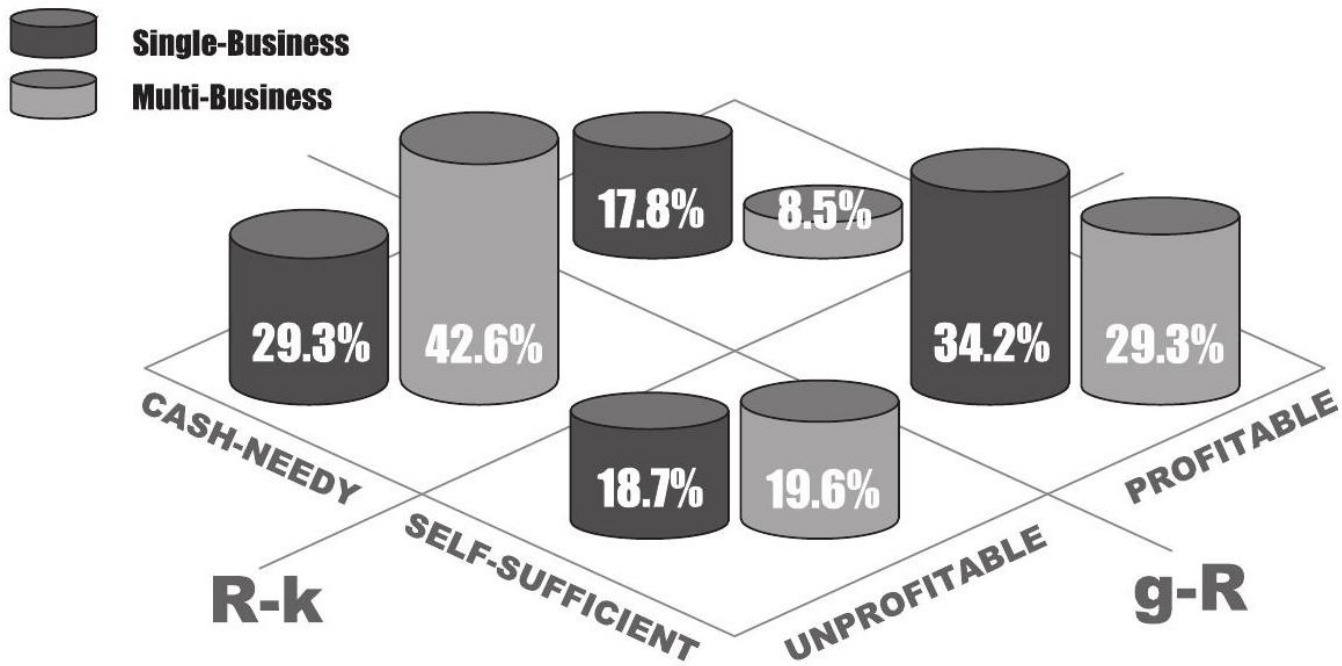

Figure 2 Location of assets by region and year. The percentages represent the relative amount of assets in each region with respect to the total assets in the sample (for single-business and multi-business firms separately.)

Decreases the investment flowing to Cash-Needy business units ($t = -3.46$).

Increases the investment flowing to Unprofitable business units (t = 4.41).

Increases the investment flowing to CNU business units ($t = 2.31$).

We also estimated a slightly different specification of the previous regression, this time using a different dummy variable for each one of the four regions (CNU, CNP, SSP and SSU) in the g-R matrix, together with their interactions with LND3. The estimated coefficients (Models 3 and 4 in Table 2) reflect the same tendency described in the previous paragraph. Again, we observe that both low-profitability regions (represented by the CNU and SSU dummy variables) present a lower level of

Table 2 Linear regression models for Capital Investment (non-financial segments only)

	Model 1		Model 2		Model 3		Model 4
	Coef.	t	Coef.	t	Coef.	t	Coef.
Intercept	0.0167	13.66	0.0228	15.85	0.0167	13.66	0.0228
SEG_G	0.0642	11.67	0.0399	9.37	0.0642	11.67	0.0399
SEG_PROF	0.0236	3.35	0.1299	17.76	0.0236	3.35	0.1299
IND_CAPINT	0.8887	19.33	0.8130	38.62	0.8887	19.33	0.8130
CASHNEEDY	0.0059	2.72	-0.0009	-0.46			
UNPROF	-0.0186	-9.81	-0.0141	-6.46			
CASHNEEDY*UNPROF	-0.0085	-3.18	-0.0069	-2.77			
CNU					-0.0212	-9.99	-0.0219
CNP					0.0059	2.72	-0.0009
SSU					-0.0186	-9.81	-0.0141
LND3	-0.0024	-1.69	-0.0057	-2.69	-0.0024	-1.69	-0.0057
LND3*CASHNEEDY	-0.0111	-3.46	-0.0069	-2.77			
LND3*UNPROF	0.0092	4.41	0.0083	3.29			
LND3$CASHNEEDY$UNPROF	0.0092	2.31	0.0045	1.3			
LND3*CNU					0.0073	2.59	0.0058
LND3*CNP					-0.0111	-3.46	-0.0069
LND3*SSU					0.0092	4.41	0.0083
Fixed effects	Industry		Time, segment		Industry		Time, segm
Years sample	1989-2004		1989-2004		1989-2004		1989-2004
Number of observations	39,452		39,452		39,452		39,452
R^2	0.172		0.103		0.172		0.103

CAPINT is the dependent variable. SEG_G and SEG_PROF are segment yearly growth and profitability, respectively. IND_CAPINT is the median level of investment intensity in the segment's industry. LND3 is the logarithm of the number of segments of the firm to which the segment belongs. CASHNEEDY equals 1 when the segment is located in either the CNU or the CNP region and 0 otherwise. UNPROF equals 1 when the segment is located in either the CNU or the SSU region and 0 otherwise. CNU, CNP and SSU identify segments located in each one of those regions. investment when considering businesses from diversified and focused firms together. The coefficient for LND3*CNU is significantly positive ($t = 2.59, P < 0.05$), which again indicates that corporate businesses in the CNU region receive more capital than stand-alones.

Success and Survival in Growth Low-profitability Businesses

Multi-business corporations appear to have more of their assets in unprofitable businesses and appear to invest more heavily in them, especially if they are Cash-Needy. Such behavior might be

aimed at future payoffs. Or, it might reflect an inefficient bias in the management of multi-business firms. In this section we attempt to takes some preliminary steps towards resolving this issue.

One could divide cash-needy unprofitable business units into two types. On one hand, there are "seeds", which are businesses that are at an initial stage of development, yielding low returns now yet holding the promise of higher profits later. In our sample, an example of this type of business is the Drilling division of the oil company Helmerich and Payne, which started as a cash-needy unprofitable business in 1989 but moved to being a cash-needy profitable business about 8 years later and then became a "cash cow" in five more years. On the other hand, there are "weeds", which are businesses that are simply underperforming. In our sample, an example of this type of business is the Cosmetic division of Del Laboratories Corporation, which is consistently found in the CNU region (cash-needy unprofitable) in a 15-year span. Naturally, corporations would be justified in investing more intensively in CNU businesses if there was a higher presence of "seeds" in corporations and, thus, the likelihood of those CNU businesses becoming profitable was higher as well.

Analyzing whether multi-business firms are more successful than single-business ones at managing Unprofitable businesses toward profitability requires an appropriate measure of what constitutes "success". We use the g-R matrix to track the evolution of the business units in our sample over time. Our analysis focuses solely on CNU businesses (CNU region). This is the region that contains "seeds" and "weeds".

We created a sub-sample of business units by taking only the observations that were located in the CNU region in 1989. We then tracked their evolution in the g-R matrix up until year 2004. In order to smooth out potential noise in the data and facilitate the analysis, we divided the period 1989-2004 into four periods of 4 years. [10] Then, we identified the business units located in the CNU region in the first period

All figures represent counts of business units except when percentages are shown. The first column shows the initial number of business units in the CNU region. The other columns show the location of the surviving business units from that initial cohort. (1989-1992) and tracked their movement in the subsequent periods. Table 3 presents this evolution.

Note that, in general, businesses that start in the CNU region in the first period have a high rate of failure over time. There were 860 business units in the CNU region in the 1st period but only 229 of those remained in the sample in the fourth period. [11] Second, we observe that a large number of business units in the CNU region tend to remain unprofitable over time. Of those starting in CNU in period 1 and surviving to period 2, 51% were still in CNU and 79% were still unprofitable. By period 3, 35.2% were still in CNU and 53% were still unprofitable. By period 4, 31.4% of the

[10] We use numerical labels (integers from 1 to 4) to label the four regions in the graph. This way, we can compute an average number for each four-year period and business unit. We round up this average to the nearest integer to come up with an "average region" in which a business unit is in over the 4 years that constitute each period.

[11] COMPUSTAT data is known to be inconsistent in its reporting of the business units within any firm. Thus,

Table 3 Evolution over time

	1989-1992	1993-1996		1997-2000		2001-2004	
All segments							
SSU		188	28.3%	74	17.8%	53	23.1%
CNU	860	340	51.2%	146	35.2%	72	31.4%
CNP		96	14.5%	90	21.7%	39	17.0%
SSP		40	6.0%	105	25.3%	65	28.4%
		664		415		229	
Diversified only							
SSU		103	30.9%	36	16.8%	30	21.7%
CNU	414	156	46.8%	69	32.2%	42	30.4%
CNP		47	14.1%	38	17.8%	25	18.1%
SSP		27	8.1%	71	33.2%	41	29.7%
		333		214		138	
Focused only							
SSU		85	25.7%	38	18.9%	23	25.3%
CNU	446	184	55.6%	77	38.3%	30	33.0%
CNP		49	14.8%	52	25.9%	14	15.4%
SSP		13	3.9%	34	16.9%	24	26.4%
		331		201		91	

survivors were still in CNU and 54% were still unprofitable.

As a rough comparison of success rates, Table 3 shows that out of 414 multi-business segments in the CNU in period 1, 156 were still in that same region (47% of the surviving units) during period 2 and 42 were still there in period 4 (30% of the total surviving units). Similarly, out of the 446 standalone business starting in the CNU region, 184 were still in that same region (56% of the total surviving units) during period 2 and 30 were still there in period 4 (33% of the survivors). These similar rates of success suggest that multi-business firms are not significantly better than single-business firms at managing underperforming business toward profitability. The results in Table 3 also suggest that CNU businesses are less likely to disappear when they are housed within a multi-business firm. Out of the business units belonging to multi-business firms located in CNU in the period 1, 33% survive until period 4. On the other hand, the corresponding survival rate in single-business firms is 20%.

A more elegant way of approaching this question is a multinomial probit model, allowing the dependent variable to take on several discrete values. Here, we label each one of the regions in the g-R matrix in the following manner: 1 for SSU, 2 for CNU, 3 for CNP and 4 for SSP. We define

some of the observations that "disappear" from the sample over the period of time analyzed might be attributable to changes in segment reporting and not to failure. For instance, a segment could be merged into another one within the same firm and that could explain its disappearance from the sample. However, our choice of defining "business units" at the three-digit level SIC code reduce this potential problem since firms often change their reporting within the four-digit level while the definition at the three-digit level remains the same.

the dependent variable FINREGION as the region numerical label (from 1 to 4) that a business unit occupies in the final period (2001-2004). We also allow FINREGION to take on value 0 if the business unit has disappeared from the sample before that final period. The multinomial probit measures the chances of any business unit in the sample occupying each one of the four regions (values $1-4$) or vanishing (value 0).

A key independent variable in our model is the dummy variable DIV, which is 1 for the business units that belong to diversified firms and 0 for the ones that belong to focused firms. A positive coefficient of DIV would suggest that being part of a diversified firm implies a higher likelihood of being in a certain region. Additional control variables are included in the model. Specifically, AVGCAXR is the average level of investment intensity (capital expenditures over total assets) for the target business unit over the whole period (1989-2004). AVGINDCAPM is the average profitability of the target business' industry over that same period and AVGINDSALE is the average growth of the target business' industry over that same period of time.

Table 4 presents the results of this probit estimation. Panel A shows the prob- ability that a business unit starts in the CNU region in period 1 and disappears from the sample before the end of period 4. The coefficient for the dummy variable DIV is significantly negative ($z = 2.55$), suggesting that being part of a multi-business firm increases the chances of survival for any given business in the CNU region. This is consistent with the view that multi-business firms tend to support a poorly performing business longer than stand-alones. Panels B, C and D show the probit estimates for the probability that a business that starts in the CNU region in period 1 ends up

FINREGION is the dependent variable and it equals 1 if the target business unit ends up in either the top-right or the bottom-left region in period 4 (2001-2004). INV is the 16-year average business unit investment ratio (capital expenditures over assets). INDPROF and INDGROWTH are the 16-year average business unit's industry yearly profitablity and growth, respectively. DIV equals 1 when the segment belongs to a diversified firm and 0 otherwise. in region SSU, CNU and CNP, respectively. We observe that the estimates for DIV are not statistically significant, thus providing no evidence that being part of a multi-business firm improves the chances to become profitable over time. [12]

Discussion and Conclusions

Chandler's pioneering work described the development of large-scale enterprise in the United States. The first wave of development exploited the efficiencies of coordinating vertical chains of operations via a centralized administration. The second wave of development came as issues of product development, differentiation, and the utilization of accumulated managerial skills came to the fore. Responding to these forces, most multi-product firms, Chandler observed adopted product-division forms of structure.

From an economic perspective, the widespread displacement of functional, or U-form structures

by product-division, or M-form structures, suggests that product-division structures are more efficient than functional structures in a multi-product context. According to Williamson (1975) this efficiency flows from organizing around the pursuit of profit by line of business rather than the functional goals of cost and revenue. And, it flows from lifting the responsibility for operational coordination from top management, freeing it to concentrate on resource allocation and strategy.

Still, if "A" is more efficient than "B", it does not follow that "A" is perfectly efficient. It might be, for example, that multi-business firms are better at operational efficiency than innovation and better at harvesting cash from mature businesses than at closing down unprofitable side lines. This study has focused on patterns of capital allocation and has identified several anomalies that may mark the presence of biases and frictions in the management of multi-business firms. In particular, we find that more diversified firms invest relatively more in unprofitable business units, less in cash-needy businesses, and more in cash-needy unprofitable businesses. What might be the sources of these anomalies? It is impossible to offer definitive conclusions from this data-it is probably impossible to offer definitive conclusions from any obtainable data on the subject. Nevertheless, having studied the question for a considerable time, we owe the reader the fruits of our reflections.

One possibility is that our study of "averages" misses the blockbusters. That is, multi-business firms might subsidize CNU businesses because, once in a while, one of them really takes off. We cannot completely discount this possibility or measure precisely the extent of this phenomenon. However, our study of the dynamics of these segments in Section VI suggests that multi-business firms are really not that successful in finding and nurturing these blockbusters.

Our results are roughly consistent with the account of "corporate socialism" developed in the corporate finance literature. Some of the work in this line (e.g. Wulf, 1999; Rajan et al., 2000; Scharfstein and Stein, 2000) stresses the agency conflict between division managers and corporate headquarters. Division managers are portrayed as rent-seeking agents that try to obtain additional compensation (in the form of extra capital allocations, among others) from corporate headquarters. They try to do so by overstating their divisions' prospects or by engaging in direct lobbying. In turn, corporate headquarters might decide that avoiding this inefficiency in resource allocation is not worth the cost of increased monitoring or low morale and thus accede to their demands. In particular, Scharfstein and Stein (2000) make the point that managers from weaker divisions have a stronger incentive to engage in firm politics given that their demands for capital investment cannot

[12] Of course, we cannot rule out the possibility that multi-business firms somehow make those CNU business units better by recombining them with other business units and achieving synergies. Karim (2006) has shown that multi-business firms in medical and pharmaceutical sectors benefit from this kind of recombining. The level of detail in our data does not allow for a clear test of that phenomenon. However, we checked whether recombining at the 3 -digit SIC level would be widespread in our data. We found that only 28 business units out of the 414 starting in the CNU region might have been recombined, simultaneously observing its disappearance and some other BU in the firm growing its assets by an amount similar to the assets of the disappearing unit. We also found only 21 CNU units that belonged to firms that saw their number of units decrease some year in the studied period but not their total assets (maximum 10% increase), which again would point at some recombining going on. Overall, recombining does not seem a widespread issue in our sample and hence it cannot be the main driver of our results. Obviously, we cannot prove that it has no effect and we find Karim's (2006) results suggestive.

be argued so effectively solely on the base of a prospect's quality. Therefore, those managers end up receiving more investment capital than they should and that creates the comparative difference with their stand-alone peers in the same industry.

The corporate socialism argument rests on a complex set of relationships among various agents within the corporation. A simpler theory is offered by the literature on cognitive biases. In particular, the allocations observed in this study can be explained as a consequence of behaviors called "naïve diversification" and "partition dependence". Naïve diversification (Benartzi and Thaler, 2001)-also known as the " $1/n$ heuristic"-is the tendency for individuals to be biased toward even allocations. Partition dependence is a consequence of naïve diversification when the decision-maker faces a particular partition of the set of choices. In the case of capital allocations, this partition of choices would be the organization of the business units within the company.

Naïve diversification and partition dependence are well-observed phenomena in other fields of human decision-making. For example, Benartzi and Thaler (2001) found a similar effect in both laboratory and field data studies of investment in 401(k) plans. When asked to choose between investing in a stock or bond fund, many individuals choose to invest 50% in each. When asked to choose between two stock funds and a bond fund, many individuals choose to spread allocations equally among the three funds, which creates an aggregate investment that is more heavily weighted (2/3) to stocks. Bardolet et al. (2009) found strong naïve diversification and partition dependence effects in managers facing hypothetical capital allocation tasks. The naïve diversification account applied to internal capital markets would predict a tendency toward equal allocation among all the business units in a firm, thus underweighting factors that would demand more uneven allocations (such as growth rates, profitability, etc). The experimental character of Bardolet, Fox and Lovallo's study shows that even in situations where social and political factors are not in play (i.e. a laboratory environment) those two biases are enough to cause a tendency toward even capital allocations among all business units, thus corporate socialism is a sufficient but not necessary explanation of inefficient allocations.

Further research on the anomalies we have identified seems warranted. In particular, it seems worthwhile to try to identify the relative importance of incentives, inertia, and biases towards even allocations in driving this result. One step in this direction would be a study which included data on corporate incentive mechanisms and changes in administration.

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Table 4 Multinomial probit

	Coeff.	St. dev.	z-value
Panel A			
Prob(FINREGION = 0)			
DIV	0.446	0.175	2.55
INV	-1.983	1.220	-1.63
INDPROF	0.121	1.761	0.07
INDGROWTH	-0.808	2.095	-0.39
cnt	1.968	0.215	9.17
Panel B			
Prob(FINREGION = 1)			
DIV	-0.105	0.225	-0.47
INV	-1.269	1.371	-0.93
INDPROF	1.823	2.086	0.87
INDGROWTH	-4.482	2.925	-1.53
cnt	0.236	0.281	0.84
Panel C			
Prob(FINREGION = 3)			
DIV	0.170	0.242	0.70
INV	2.503	1.170	2.14
INDPROF	1.085	2.013	0.54
INDGROWTH	4.184	2.608	1.60
cnt	-0.833	0.295	-2.83
Panel D			
Prob(FINREGION = 4)			
DIV	0.134	0.218	0.62
INV	1.537	1.099	1.40
INDPROF	5.823	2.072	2.81
INDGROWTH	2.503	2.516	0.99
cnt	-0.272	0.263	-1.03
Number of observations	860		
Wald chi ^{2} (16)	48.36		
Prob $>$ chi ^{2}	0.000		